Early ICU Sepsis Prediction with SEPSIS SENTINEL MODEL

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ABSTRACT

Early detection of sepsis in the ICU is challenging due to heterogeneous patient deterioration and low event prevalence. We developed an ICU-hour–level prediction model using aggregated vital signs to forecast sepsis onset within a 1–4 hour window.

DATASET & FEATURE ENGINEERING PIPELINE

MIMIC-IV ICU

26k ICU stays
940k hourly records
4.5% sepsis prevalence

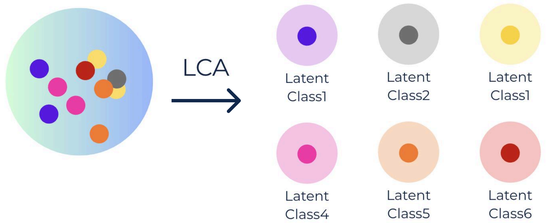
Cohort Extraction

SQL cohort extraction
ICU index time

Hourly Feature Engineering

Hourly time-series features
Vitals
Labs
SOFA
Demographics

Latent Multimorbidity Phenotypes

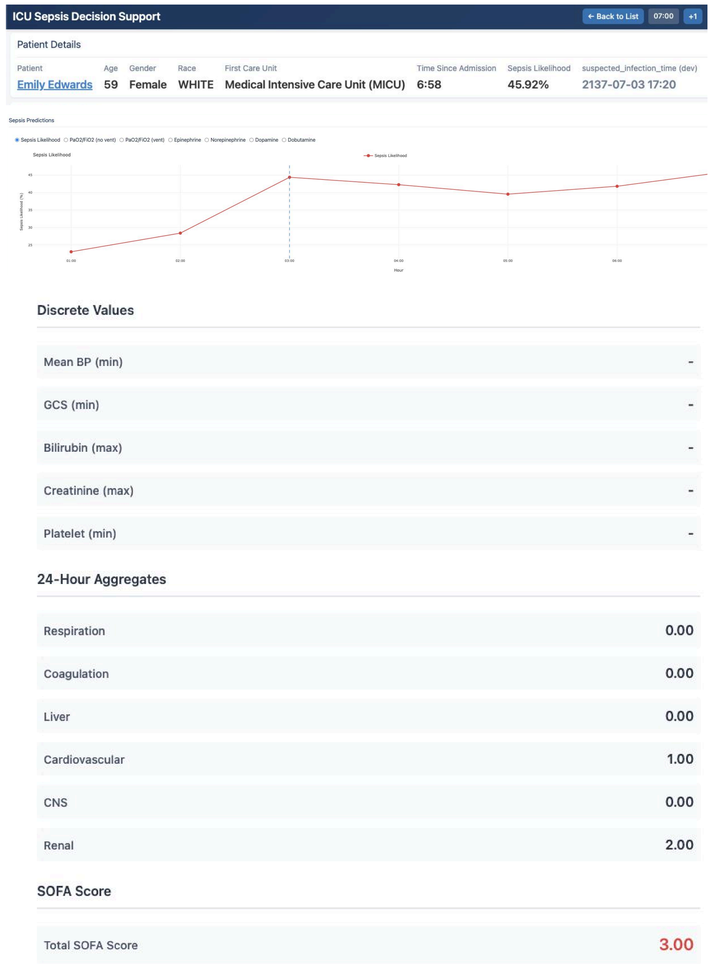


- Cardiopulmonary
- Cardiac
- Young
- Hepatic/Addiction
- Complicated Diabetes
- Uncomplicated Diabetes

Model Input

Hourly features
Sepsis label
Subtype

CLINICAL APPLICATION



SEPSIS SENTINEL MODEL

Model Input

Hourly features
Comorbidity subgroups
Sepsis lable(y)

Patient Split

80% train
20% Test

GroupShuffleSplit
Group = Stay_id

Preprocessing Pipeline

Numeric(Median Impute)
Categorical(One-Hot Encode)

Column Transformer

Random Forest Classifier

500 Trees
max_depth=10

Risk Prediction

Sepsis risk probability
$P(\text{sepsis} \mid X) \in [0,1]$

Model Deployment

API inference service
Sepsis risk output

MODEL PERFORMANCE

Cost Matrix

	Predict No Sepsis	Predict Sepsis
Sepsis Negative	53,224	68,471
Sepsis Positive	17,860	278,381

The model prioritizes minimizing false negatives to improve early sepsis detection.

Overall Performance

Metric	Value
Accuracy	0.79
Balanced Accuracy	0.69
Specificity	0.44
FPR	0.56
NPV	0.75
Sensitivity	0.94
Precision	0.80
F1 Score	0.87
FNR	0.06

CONCLUSION

Our ICU-hour sepsis early warning model using vital signs from the MIMIC-IV database. The model predicts sepsis 1–4 hours before onset, achieving 81% sensitivity and 95.9% specificity, with alerts enriched more than 10× over baseline prevalence. Integrating multimorbidity context captures patient heterogeneity and improves clinically actionable ICU risk prediction.